

CYNX 1103 Official Writeup

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Difficulty: Medium

Writeup classification: Official

1 – Introduction

CYNX 1103 is a medium cryptography challenge which challenges the problem solving ability of players. The title is a direct flip of the famous `Cicada 3301` which is also a hint to the challenge itself. The player must identify the Gematria Primus encoded runes which include a Vigenere layer as well as a Shift-cipher layer to retrieve the keys needed to obtain the flag.

2 – Description

A simple note reads:

“Do not mistake the butterfly for a moth”

-0xCipher

3 – Skills Required

- Familiar with standard cryptography encryptions
- Familiar with runic ciphers
- Familiar with Cicada 3301 style puzzles
- Knowledge of scripting for custom decoding patterns

4 – Skills Learnt

- Learn about the Gematria Primus encoding
- Learn how to script for custom vigenere/shift cipher decoding

5 – Solve

Connecting to the port, the user will be greeted with this interface:

```
(kali㉿kali)-[~/Desktop]
$ nc 145.79.11.93 9000

 6 6      .6 66      66 6.      6 6
 6 6 *6,      66 (6,      66      66      66* 66      6. 6 (6
 6 666      66 6% 6 6      6      6      6 6 66 66      .666 6
6666 66666%/(6666 6666 6 6      6      6      6 6 6666 6666(*%66666 6666
66% 66666/      6      66 6      * %      6 66      6      (66666 666
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6666      66 6      66 6      6 66      6666      6666
6 6 6 6 6,      6 6.      6 6*666, 6      .6 6      .6 6 6 6
6 66666      66      .66      66      66
6 6/66, .6.      6 6 66.6 /      .6, 66,6 6
6 6666, *6666 6.66      6 / /6 .6 * 6, 66.6 6666, *6666,6
66666 %*      /6 666(6.666 6(      ,6 66666
6666%      6      6 666      6      66666
6, 66 6,      *      6 666      6      , 6 6 ,6
666%      6      66 66      6      6666
6* .66 /      6 6      6, 6.6 6      6 6 / 66 *6
66666 6      (      %      6* 6      ,      6 6666%
( 6666      6      6      66 6      6      6666 /
6 6 6666      . 6, 6      6 6      6666 6 6
. % 6 66 66 666.6      6/666 66 66 6 6 /
6 6( 66(.6 %6666      66666 6*66 .6 6
.66 66 6      6 /6 66
66      66
/6,      %6

0xC1ph3r's Vault
1. Get key #1
2. Get key #2
3. Get key #3
4. Submit keys
0. Quit
Choice: █
```

If the user selects option 1-3, they will be given a block of encoded text:

```
Choice: 1

PMLEMM

PMLEMM PFFT TF BM EFB FP M5IXN0MM5M5T
BM RFFW TF XRMFT5M44 14 FFM BMFFRM AFD

MF 5FT TRMMBIM 5FR MF AFD MFD5T
PFR AFR P1MM4 RMMF5 55TF5NMM F5W PNF5M

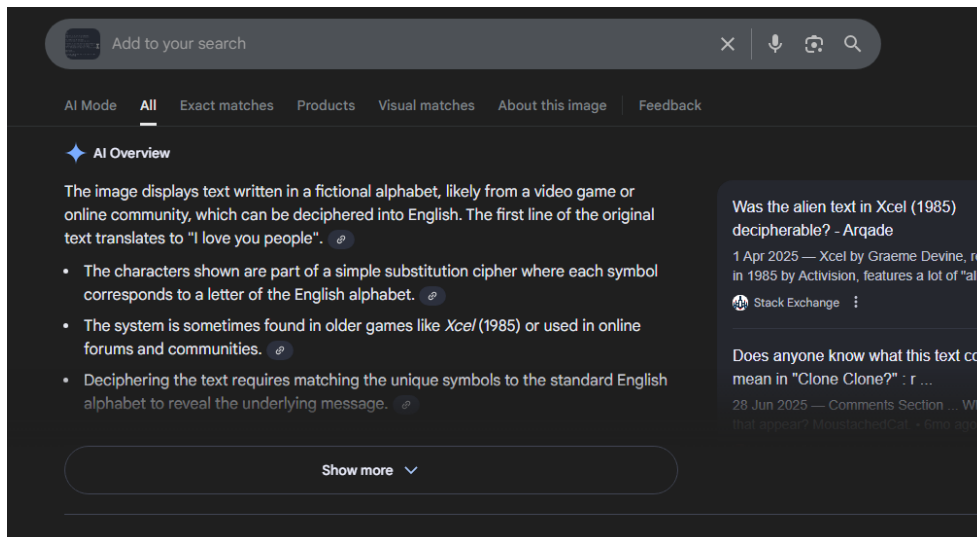
PNE5 PM 4MML 14 5FT AFR PMFR 5FR AFR RM414TF5LM
BD5T F 4MFT5 PR5XMM5T FP AFR TIMM PRMM5X 51DM5
FR 15MM5TFB5A TFLM5 514 EFB 14 5FT 4NFR5T
5FR 14 1T 5M555M 5M5T F5T PNF PFL 1T FRR5M TRF54PF5 MMW

FBF5WF5 NF4TM
FBF5WF5 LMR5F55A

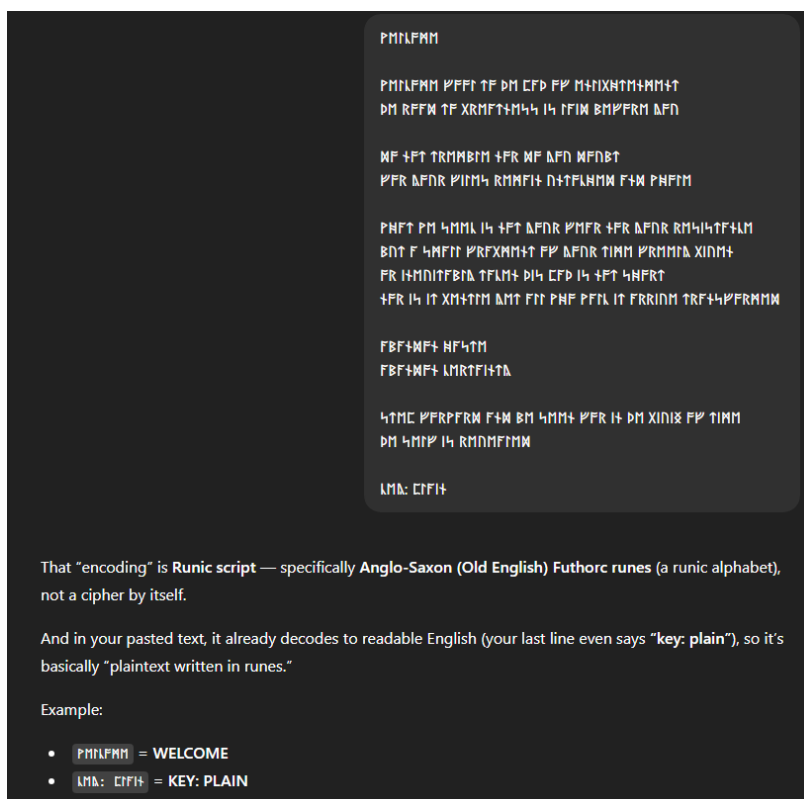
4TME PFRPFRW F5W BM 4MM5 PFR 14 5M 51N1X FP TIMM
5M 4MT5 14 RMMMF5M

LMA: EFT5
```

If the user attempts to reverse image search this block of text, they will be greeted with either a completely wrong answer or a slight red herring:



AI however, will be able to solve the simpler parts of the key texts:



But still, Anglo-Saxon is still the wrong `runic` encoding. This encoding can work for the most part but will start giving wrong decoded texts for some other parts.

Regardless, the correct encoding for this is the Cicada 3301's Gematria Primus encoding.

With that, the user can search for this [specific tool](#) that exists to decode the text:

TRANSLATE WITH GEMATRIA PRIMUS

RUNES TO TEXT

FULL LIBER PRIMUS

REMOVE LINE BREAKS

FIND WORD PATTERN

WORD LENGTH (E.G. DIVINITY = 8)

REPEATED LETTER POSITIONS (E.G. DIVINITY = 1,3,5)

FBF+WF+ LMRTF+TA

4TMC PFRPFRM F+M BM 4MM+ PFR I+ BM XINIX FP TIMM

BM 4MTF I+ RMNMFTMM

LM: CTF+

TRANSLATE

RESULT:

STEP FORWARD AND BE SEEN FOR IN THE GIVING OF TIME
THE SELF IS REVEALED

CEY: PLAIN

Key #1 = PLAIN

For the 2nd wall of text, the encoding only works until right before the key:

Choice: 2

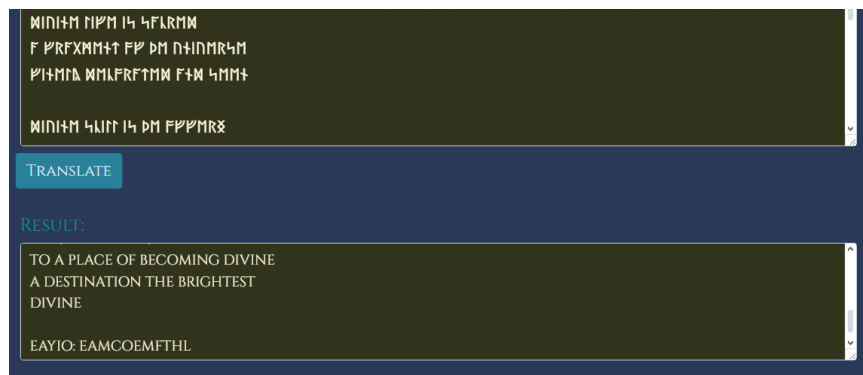
4FP I+ XMTH F TITTTM WINDISM
4F+X WINDISM I+ XINMT PFR PRMM
WINDISM TIRM I+ 4FLRMM
F PREXMM+T FP BM N+INDR4M
P+MM+ WMLFRFMM F+M 4MM+

WINDISM 4LIT+ I+ BM FPPMRX
LFRMM I+ TIMM F+M 4TRIPM
WINDISM NF+MM BA F44 F+M 4TRNXXIM

WINDISM 4NFCMM BA TIRM
WINDISM I+ PITT TFLM WFN 4FMM+MMRM

BM4F+M BM 4NFWFP
BM4F+M BM 4IXNT
TF F CTFM FP BM4F+X WINDISM
F WMYT+FTIF+ BM BRXNTMT
WINDISM

TTX: 4MFTMMTX



What's going on?

Looking closer at the script itself, the user is able to see a frequent repeat of the text
`DIVINE`:

NOW IT GETS A LITTLE DIVINE
NOTHING DIVINE IS GIVEN FOR FREE
DIVINE LIFE IS SACRED
A FRAGMENT OF THE VNIVERSE
FINELY DECORATED AND SEEN

DIVINE SCILL IS THE OFFERING
CARVED IN TIME AND STRIFE
DIVINE HONED BY LOSS AND STRVGGLE

DIVINE SHAPED BY LIFE
DIVINE IT WILL TACE YOY SOMEWHERE

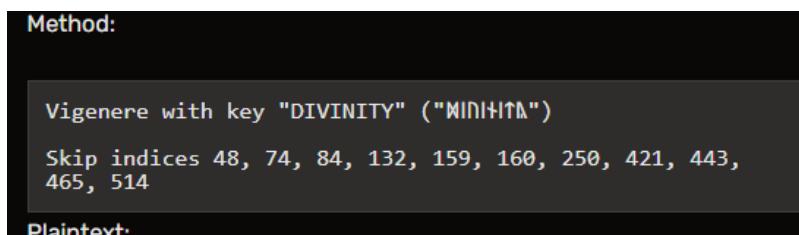
BEYOND THE SHADOW
BEYOND THE NIGHT
TO A PLACE OF BECOMING DIVINE
A DESTINATION THE BRIGHTEST
DIVINE

EAYIO: EAMCOEMFTHL

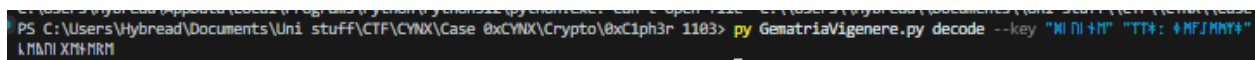
With this information, the user can assume that the key has been encrypted with Vigenere.



Why isn't it working? Doing a little extra research, the player can find the needed information in this page: https://uncovering-cicada.fandom.com/wiki/How_the_solved_pages_of_the_Liber_Primus_were_solved



This tells us that a custom Vigenere encryption was taken place where the rune itself is encrypted with the word `DIVINE` (`MINIHTM`). To perform this decryption, the user will need to craft a custom Vigenere decryption script which uses the Gematria Primus index rune mapping instead of the standard A-Z:



LMANIXH+MRH

TRANSLATE

RESULT:

CEYVIGENERE

Key #2: VIGENERE

Finally, for the 3rd key, we're provided with another wall of text which, this time, doesn't decode properly:

WORD LENGTH (E.G: DIVINITY = 8)

REPEATED LETTER POSITIONS (E.G: DIVINITY = 1,3,5)

PLAINTEXT

R*T! 8 FMMHIAIAR* X*IAI AI XIRMX
ITM MJI I CRMX *RI BRXM
M*TAHTIM*BIM*I AX *RI PRX*E
AI JFFAXMX WTM* 8 PRFNMIX AXMITP
DMI: XTAIP

TRANSLATE

RESULT:

LEAE WYIOEOT DEOLEA LR EIOTYVEALEIOBEIOL
8 ETHEAREING MEWROE BEEIOYIOV
INGYTEIOTHE WRTPIG YIOLR 8 THROIOEOING RW LEARGVEAL
EO BYIOP INGDTYL EOIOP OETHRBMYOEP YIOLR 8

Looking closer, the user can see, similar to key #2, a common occurrence of the number `8`.

With this information, the user can conclude that it might be something like a simple Caesar Shift Cipher of 8. But again, it's a custom mapping of the shift as the user will soon find out that a copy pasting the decoded text does not work:

e.g. type 'sudoku'

BROWSE THE FULL dCODE TOOLS LIST

Results

Caesar Cipher - Shift by 8

I, J, K, L, M, N, ... G, H

A, B, C, D, E, F, ... Y, Z

DWSW OQAGWGL VWGDWS

DJ WAGLQNWSDWAGTWAGD

8 WLZWSJWAFY EWOJGW

TWVGAGQAGN

AFYQLWAGLZW OJLHAFY

QAGDJ 8 LZJGAGWGAFY

JO DWSJYNWSD

WG TQAGH AFYVLQD

WGAGH GWLZJTEQAGWH

QAGDJ 8

DGYDWS OGWGLZDYGWAFY

GWQJGTAFY

OGWGLZDYGWAFY

WGNWGQAG

8 AGJD WGWSWWGH JG

EWWSQAGH

EGWWGDWS LJJVAFY

DWSGJYNWS 8 DYGAGAFY

JO AFYDQLLAGWAFYAFY

TWTJGA HGQVAFY

DWSGJYNWS 8

YAGTWGGLZWH HJJGAFY

DWSW AFYWLO DWSQAGAFY

AFYDQWFI 7WSWH

→ 8 (←18)

CAESAR CIPHER DECODER

CAESAR SHIFTED CIPHERTEXT

LEAE DEOLEA PREING IORL BRGE

EIOTYVEALEIOBEIDL YING IORL WRGIOP

YL EOOOYGEING FEAEIO 8 WROVELING YLINGETW

THEI: INGEAYWL

Test all possible shifts (26-letter alphabet A-Z)

▶ DECRYPT (BRUTEFORCE)

MANUAL DECRYPTION AND PARAMETERS

SHIFT/KEY (NUMBER): 8

USE THE ENGLISH ALPHABET (26 LETTERS FROM A TO Z)

USE THE ENGLISH ALPHABET AND ALSO SHIFT THE DIGITS 0-9

USE THE LATIN ALPHABET IN THE TIME OF CAESAR (23 LETTERS, NO J, U OR W)

USE THE ASCII TABLE (0-127) AS ALPHABET

USE A CUSTOM ALPHABET (A-Z0-9 CHARS ONLY)

0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ

▶ DECRYPT

See also: ROT Cipher – Shift Cipher

CAESAR ENCODER

CAESAR CODE PLAIN TEXT

Smwvpoy{Q1mpr1}

(Definition

How to cipher?

How to cipher?

How to ciphertext?

How to without kn

Is the C

What ar Caesar cip

How to numbers u

Why the Cipher?

What is

What ar names?

How to the Caesar

How to in pseudo-

When C invented?

Similar

ROT Cij

Shift Cij

Vigener

To achieve the proper decryption, the user again, will have to craft a custom shift and decoding script to retrieve the actual values:

```

PS C:\Users\Hybread\Documents\Uni stuff\CTF\CYNK\Case 0xCYNK\Crypto\0xC1ph3r 1103> py GematriaShift.py
Mode (e=encode / d=decode): d
SHIFT (rotation 0-28): 8

Paste your block. End with Ctrl-D (Linux/macOS) or Ctrl-Z then Enter (Windows).

ITM PAJIT MJIT IR MTANTIMBM#
8 NDTIME MNPFRM BNF#A#N
8ATM#PM PRICE A#R 8 DRF#MFE RP ITXONT

I B#C 8MT# J#C FMDRBM#MC A#R 8
ITXIT PFJDIXFM# FMPRF# PFJDIXFM# JN/A#
8 #RT JTM#C RF MMT#C

MFMIT TRR# ITXONT 8 IXF# RP 8IATT#M#
BMBRFI CF# ITXONT 8 X#BJF#C CRF#
ITM 8MTP IT# 8I FMT#TMC JDFR# 8 BR#M#

#R 8IFT #R PA#A#Y
R#TI 8 FMM#A#R# X#A#T A# 8IR#
ITM MJIT CR# #RT BROM
MTANTIMBM# A# #RT PRO#C
A# JFFAX# PT# 8 PRFOM# A#MTP

D#I : 8TAP#
^Z
Decode output style: (c=compact / s=spaced tokens): c

Decoded (decrypted) latin tokens:

THE FINAL PATH TO ENLIGHTENMENT
8 ECHOES BEFORE MEANING
SILENCE FOLDS INTO 8 CORNERS OF THOUGHT

A MIND SPLIT AND RECOMBINED INTO 8
TRVTH FRACTVRES REFORMS FRACTVRES AGAIN
8 NOT AHEAD OR BEHIND

BREATH LOOPS THROUGH 8 TVRNS OF STILLNESS
MEMORY DRIPS THROUGH 8 VMARCED DOORS
THE SELF THINKS STRETCHED ACROSS 8 MOMENTS

NO START NO FINISH
ONLY 8 REPETITION UNTIL IT STOPS
THE PATH DOES NOT MOVE
ENLIGHTENMENT IS NOT FOUND
IT ARRIVES WHEN 8 FORGETS ITSELF

CEY: SHIFT
PS C:\Users\Hybread\Documents\Uni stuff\CTF\CYNK\Case 0xCYNK\Crypto\0xC1ph3r 1103>

```

Key #3 = SHIFT

Now with the 3 keys obtained, the user can submit it into the program:

In the following page, the user can also check whether their keys are valid or not:

```
Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
0. Back
Choice: 1
Enter key #1: PLAIN
Saved.

Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
0. Back
Choice: 2
Enter key #2: Vigenere
Saved.

Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
0. Back
Choice: 4
Key #1: Valid Key (submitted)
Key #2: Wrong Key (submitted)
Key #3: Wrong Key (not submitted)
```

With the proper keys submitted, a 5th option will appear:

```
Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
0. Back
Choice: 3
Enter key #3: SHIFT
Saved.

Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
5. Claim file
0. Back
Choice: █
```

Selecting that option will give us the following:

```
Submit keys of 0xC1ph3r
1. Submit key #1
2. Submit key #2
3. Submit key #3
4. Check key validity
5. Claim file
0. Back
Choice: 5

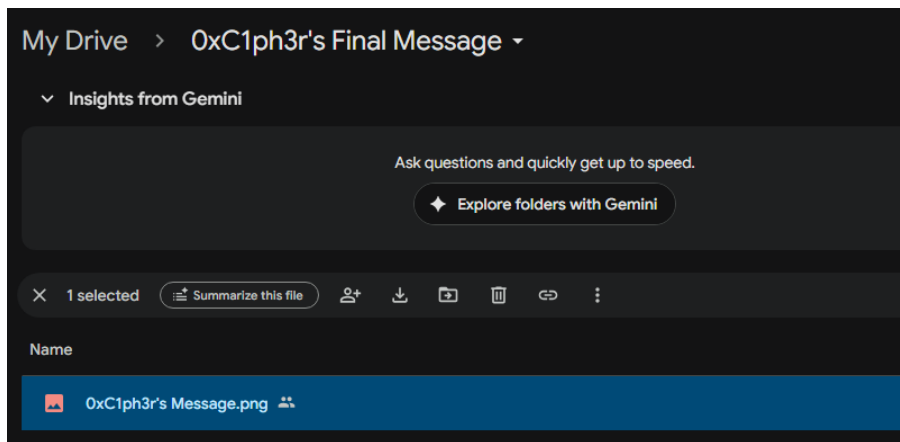
I

drivegooglecomdrivefolders1SEP1ECj8odVm-BKUctK1PMfztdaNAwXu
```

A google drive link but it needs some HTML formatting:

<https://drive.google.com/drive/folders/1SEP1ECj8odVm-BKUctK1PMfztdaNAwXu>

After fixing it, the user can head to the drive and find the following:



የ ልማተር የተሰጠው የሥራ ስልጣን
በዚህ ሰዓት ብቻ ነው የሚሰጠው
ሆኖ የሚታወቅ የሥራ ስልጣን
በዚህ ሰዓት ብቻ ነው የሚሰጠው
ሆኖ የሚታወቅ የሥራ ስልጣን
በዚህ ሰዓት ብቻ ነው የሚሰጠው
ሆኖ የሚታወቅ የሥራ ስልጣን
በዚህ ሰዓት ብቻ ነው የሚሰጠው
ሆኖ የሚታወቅ የሥራ ስልጣን

የሆነ

በዚህ ሰዓት ብቻ ነው የሚሰጠው

OxC1ph3r

A final message from OxCipher and upon decoding it, the user will obtain the flag:

FLAG
DIDYOVJVSTCRACCAPAGEOFTHELIBERPRIMVS

Flag = CYNX{DIDYOVJVSTCRACCAPAGEOFTHELIBERPRIMVS}